



Total Marks: 60 (20×3)

Time: **90 Minutes**

- Total no of questions are **four (4)**. Answer any **three (3)**.
- Figure in bracket [] next to each question indicates marks for that question.
- Write your set number on top of your answer sheet
- Use activation voltage $V_\gamma(\text{diode}) = 0.6 \text{ V}$, $V_\gamma(\text{transistor}) = 0.5 \text{ V}$, $V_{BE}(\text{forward active}) = 0.7 \text{ V}$ and $V_D(\text{conducting voltage of Diode}) = 0.7 \text{ V}$, for all the questions.
- There is no need to draw circuits in your answer sheet.

Name:	ID:	Section:
-------	-----	----------

Question 1

For the square wave generator in **figure 1**, answer the following questions.

[20]

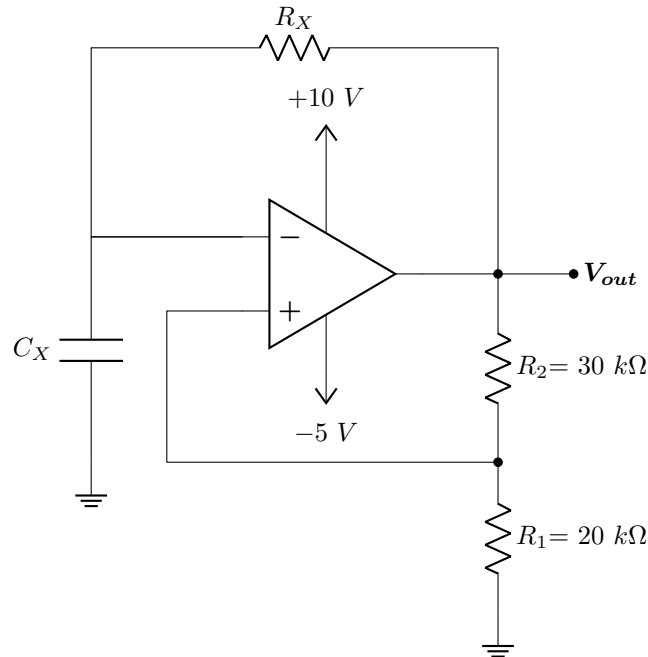


Figure 1

CO4	(a)	Identity whether the Schmitt Trigger in figure 1 is inverting or non-inverting .	[2]
CO4	(b)	Determine V_{TH} and V_{TL} values of the Schmitt Trigger. (<i>high and low threshold values</i>)	[4+4]
CO5	(c)	Determine the duty cycle of the square wave generator circuit. (<i>hint: R_x and C_x values are not required for this part</i>)	[5]
CO5	(d)	Find the value for C_x so that the circuit has a frequency of 100 Hz when $R_x = 558 \Omega$	[5]

Question 2

Consider the ECL circuit in the **figure 2** where v_{O1} represents buffer output and v_{O2} represents inverter output for the input voltage v_{in} . Ignore the base currents and suppose **logical Low** voltage is -1.8 V . [20]

CO1	(a)	Determine the value of i_E [Hint: consider v_{in} is logically Low and so Q_R is on and Q_1 is off].	[6]
CO2	(b)	If v_{in} is logic High, Determine the value of R_{C1} .	[6]
CO2	(c)	Find out the value of Reference voltage V_R [Hint: Reference voltage is the average of logic Low and logic High values]	[4]
CO1	(d)	Calculate the total power dissipated in the circuit if v_{in} is Low .	[4]

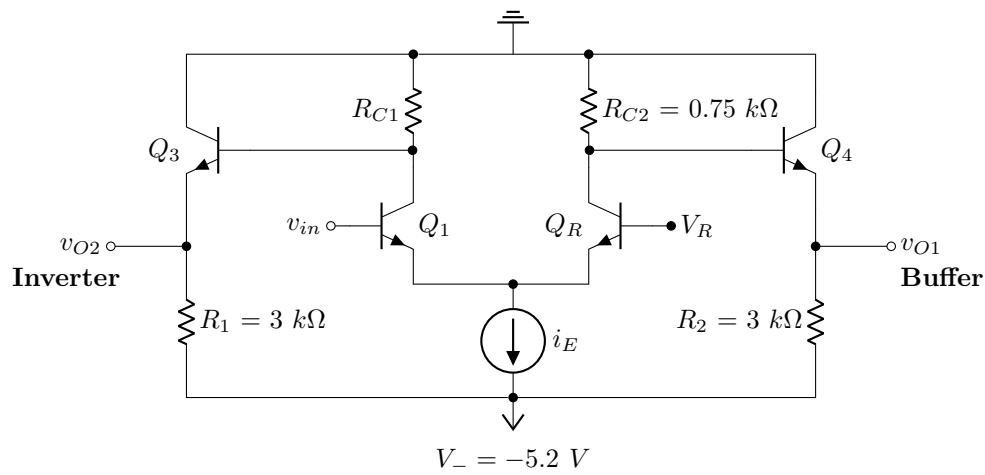


Figure 2

Question 3

Flash ADC in figure 4 has reference voltage of $V_{REF} = 16\text{ V}$. The output (V_o) of the DAC in figure 3 is used as input (V_{in}) of the Flash ADC in figure 4. [20]

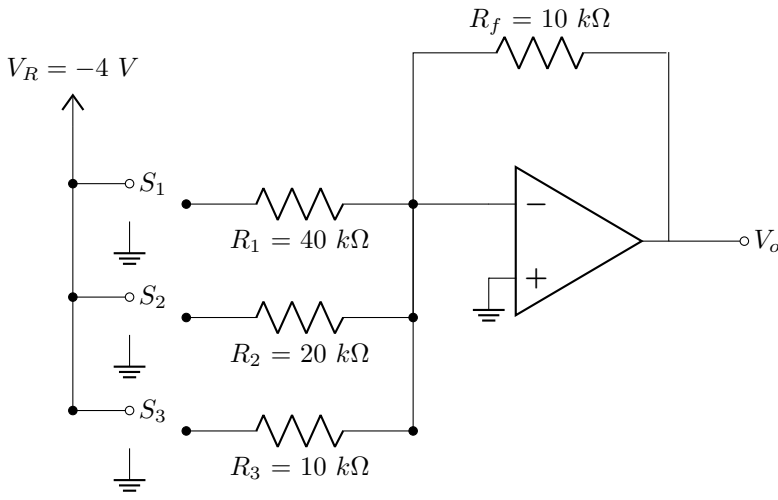


Figure 3: Binary weighted DAC

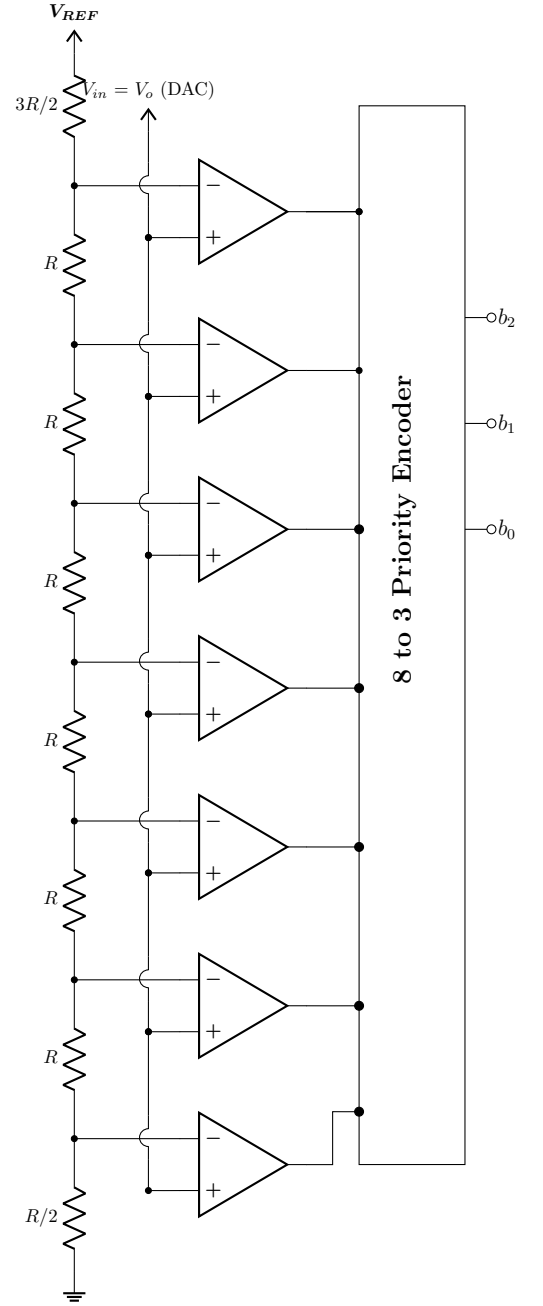


Figure 4: Flash ADC

CO2	(a)	Determine the value of V_o of the DAC for input combinations 001 and 010.	[5]
CO3	(b)	(i) Suppose the Flash ADC in the figure 4 has an output of 010. Find the range of input voltage V_{in} that produces this output. (ii) Determine all possible input combinations for the DAC that will generate V_{in} (V_o of DAC) within this range.	[6+4]
CO3	(c)	We wish to design a 2-bit flash ADC similar to the figure above to convert an analog signal. Calculate how many resistors and comparators will be required.	[5]

Question 4

For the given TTL circuit in figure 5, **common emitter current gain $\beta_F = 30$** , $V_{BE(sat)} = 0.8V$, $V_{CE(sat)} = 0.1V$, and **reverse common emitter current gain of the transistors $\beta_R = 0.1$** . Assume all inputs are **logical High** or **6V**. [20]

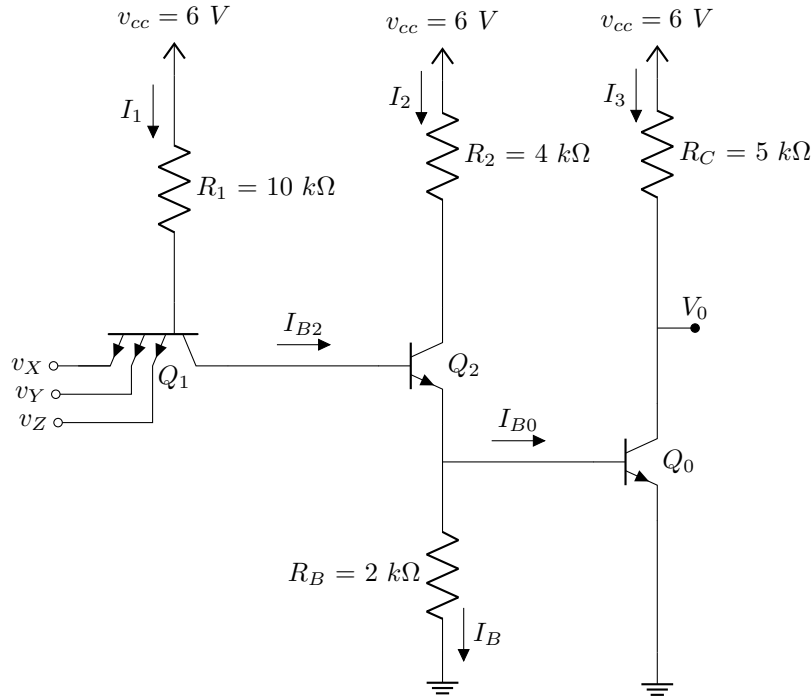


Figure 5

CO1	(a)	Calculate all the currents labeled in the above circuit.	[10]
CO2	(b)	Design a CMOS logic circuit to implement the logic function, $Y = AC + B$. Assume that the signals A, B, and C are available. (You must show both PMOS and NMOS networks).	[10]



BRAC UNIVERSITY
Department of Computer Science Engineering
FINAL EXAMINATION, SPRING 2023
CSE 350: Digital Electronics and Pulse Technique



Total Marks: 60 (20 × 3)

Time: **90 Minutes**

- Total no of questions are **four (4)**. Answer any **three (3)**.
- Figure in bracket [] next to each question indicates marks for that question.
- Write your set number on top of your answer sheet
- Use activation voltage $V_\gamma(\text{diode}) = 0.6 \text{ V}$, $V_\gamma(\text{transistor}) = 0.5 \text{ V}$, $V_{BE}(\text{forward active}) = 0.7 \text{ V}$ and $V_D(\text{conducting voltage of Diode}) = 0.7 \text{ V}$, for all the questions.
- There is no need to draw circuits in your answer sheet.

Name:	ID:	Section:
-------	-----	----------

Question 1

Consider the ECL circuit in the **figure 1** where v_{O1} represents buffer output and v_{O2} represents inverter output for the input voltage v_{in} . Ignore the base currents and suppose **logical Low** voltage is -1.8 V . [20]

CO1	(a)	Determine the value of i_E [Hint: consider v_{in} is logically High and so Q_R is off and Q_1 is on].	[6]
CO2	(b)	If v_{in} is logic Low, Determine the value of R_{c2} .	[6]
CO2	(c)	Find out the value of Reference voltage V_R [Hint: Reference voltage is the average of logic Low and logic High values]	[4]
CO2	(d)	Calculate the total power dissipated in the circuit if v_{in} is High .	[4]

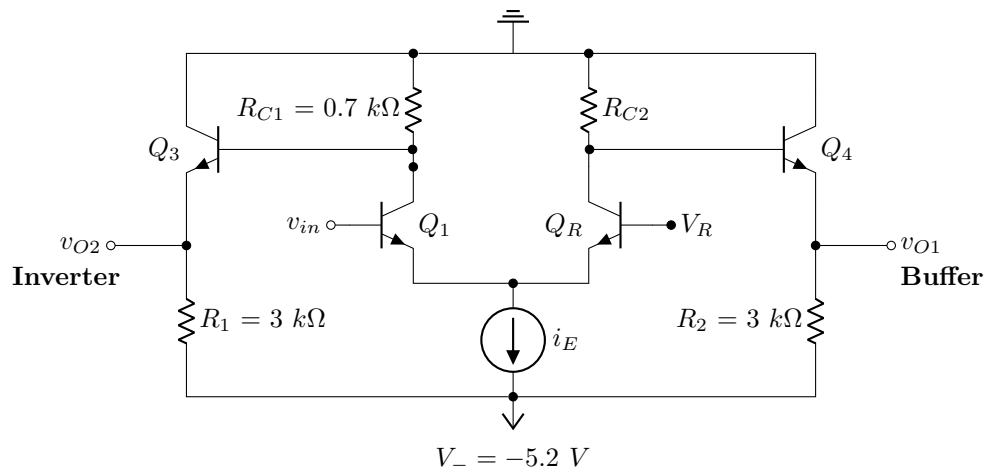


Figure 1

Question 2

Flash ADC in figure 3 has reference voltage of $V_{REF} = 32\text{ V}$. The output (V_o) of the DAC in figure 2 is used as input (V_{in}) of the Flash ADC in figure 3. [20]

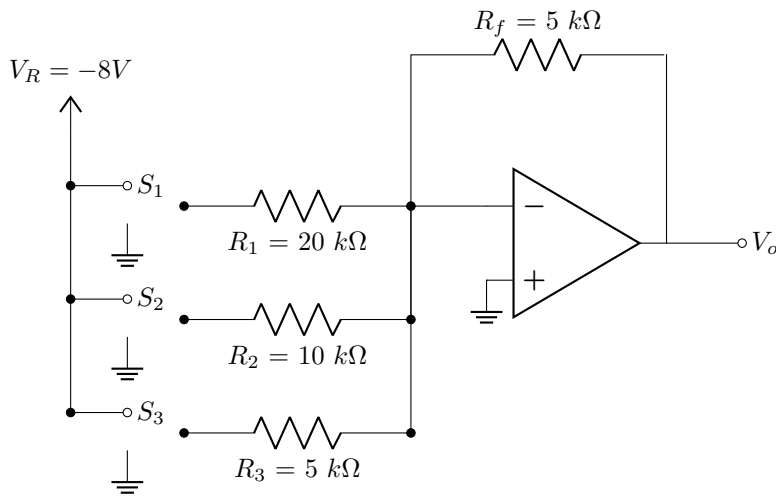


Figure 2: Binary weighted DAC

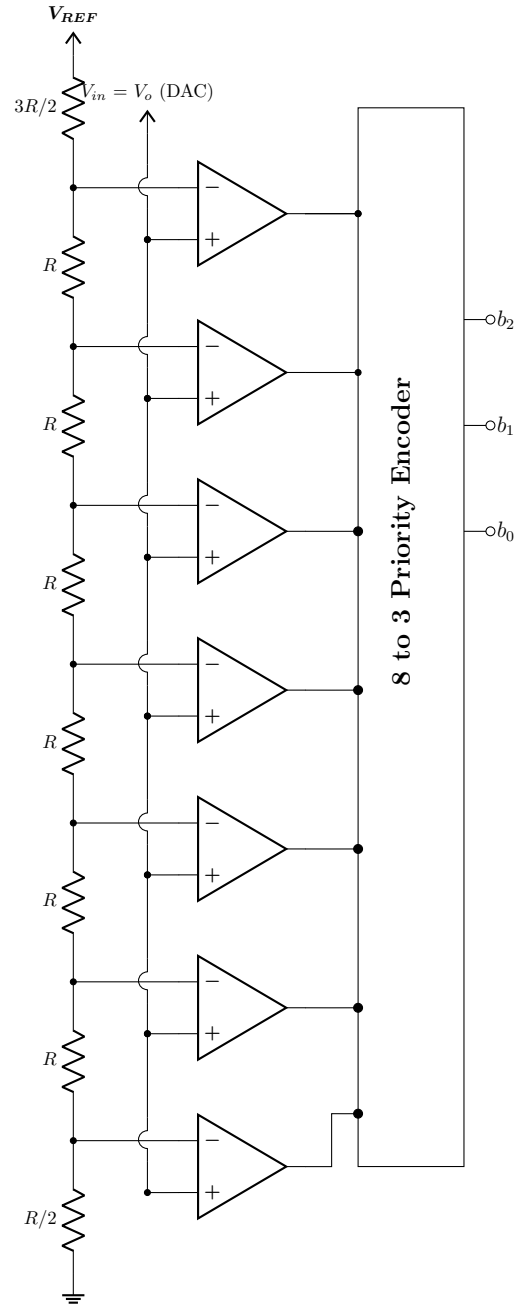


Figure 3: Flash ADC

CO2	(a)	Determine the value of V_o of the DAC for input combinations 001 and 010 .	[5]
CO3	(b)	(i) Suppose the Flash ADC in the figure 3 above has an output of 011 . Find the range of input voltage V_{in} that produces this output. (ii) Determine all possible input combinations for the DAC that will generate V_{in} (V_o of DAC) within this range.	[6+4]
CO3	(c)	We wish to design a 2-bit flash ADC similar to the figure above to convert an analog signal. Calculate how many resistors and comparators will be required.	[5]

Question 3

For the given TTL circuit in figure 4, common emitter current gain $\beta_F = 30$, $V_{BE(sat)} = 0.8V$, $V_{CE(sat)} = 0.1V$, and reverse common emitter current gain of the transistors $\beta_R = 0.1$. Assume both inputs are logical High or $6V$. [20]

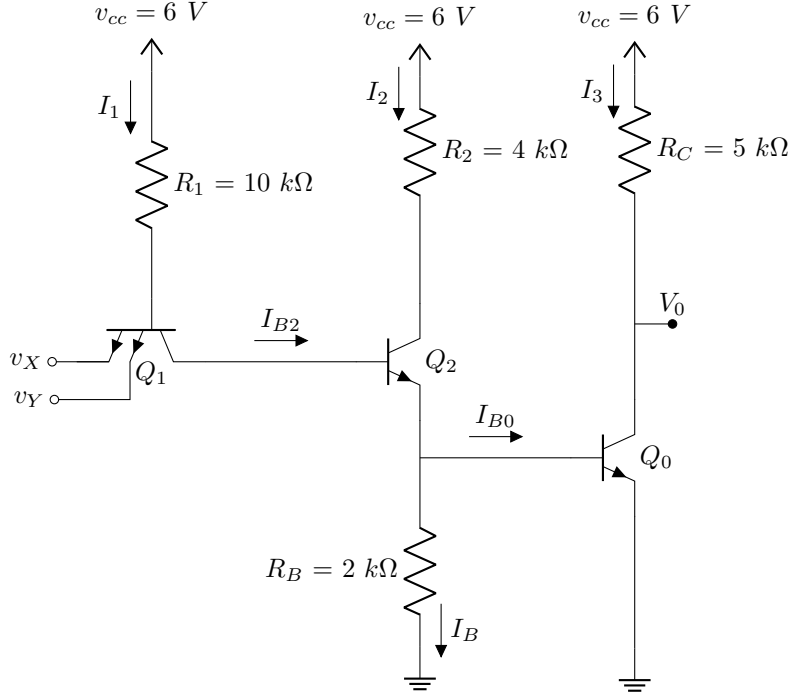


Figure 4

CO1	(a)	Calculate all the currents labeled in the above circuit.	[10]
CO2	(b)	Design a CMOS logic circuit to implement the logic function, $Y = AC + B$. Assume that the signals A, B, and C are available. (You must show both PMOS and NMOS networks).	[10]

Question 4

For the rectangular wave generator in figure 5, answer the following questions.

[20]

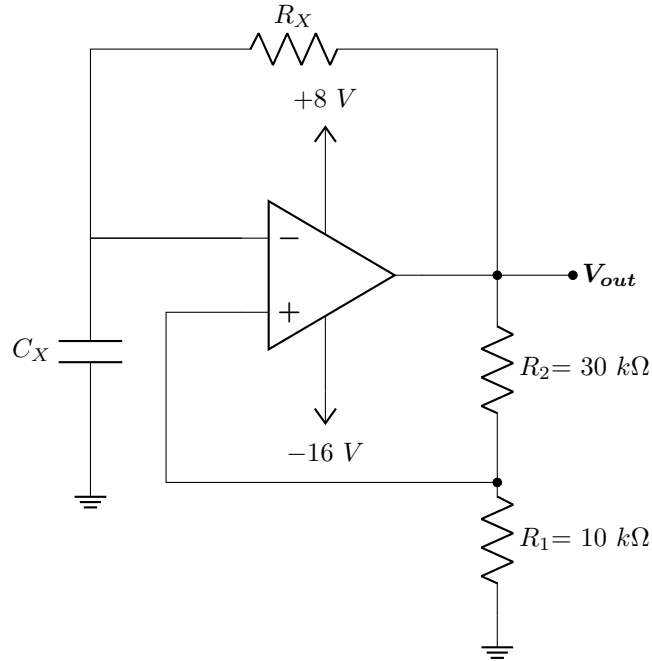


Figure 5

CO2	(a)	Identify whether the Schmitt Trigger in this circuit is inverting or non-inverting	[2]
CO2	(b)	Determine V_{TH} and V_{TL} values of the Schmitt Trigger. (<i>high and low threshold values</i>)	[4+4]
CO2	(c)	Determine the duty cycle of the rectangular wave generator circuit. (<i>hint: R_x and C_x values are not required for this part</i>)	[5]
CO2	(d)	Find the value for C_x so that the circuit has a frequency of 1000 Hz when $R_x = 910\text{ }\Omega$	[5]